

CVT System Model Derivation

Kai Arseneau

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1 Overview

1.1 Purpose of This Document

This document presents a complete dynamic derivation of a continuously variable transmission (CVT) system model. The goal is to develop a physically consistent set of governing equations that describe:

- Rotational dynamics of the primary and secondary assemblies,
- Axial shift dynamics of the primary sheaves,
- Belt traction and torque transfer,
- Slip behavior under torque saturation,
- Coupled vehicle longitudinal response.

The resulting formulation provides a structured, state-based model suitable for numerical simulation and performance analysis.

1.2 Scope

This document focuses on the mechanical modeling of the CVT system. It includes force balances, torque transfer mechanisms, and dynamic coupling between rotational and axial motion.

The following are outside the scope of this document:

- Thermal modeling,
- Long-term wear or material degradation,
- Detailed tribological contact mechanics,
- Multi-body vehicle dynamics beyond one-dimensional motion.

All assumptions limiting model fidelity are formally defined in [Section 5](#).

1.3 Expected Background Knowledge

The reader is expected to be familiar with:

- Newton’s laws of motion,
- Basic rigid-body rotational dynamics,
- Linear spring models,
- Gear ratios and torque multiplication,
- Basic algebra and differential equations.

Prior familiarity with state-space modeling or numerical integration is not assumed. Relevant concepts are introduced where required.

1.4 Conceptual Overview of a CVT

A continuously variable transmission (CVT) transmits torque between an engine and drivetrain using a belt or chain running between two variable-diameter pulleys.

Each pulley consists of two opposing conical faces (sheaves) whose axial separation determines the effective belt radius. As the primary sheaves shift axially, the effective radius changes, which in turn modifies the transmission ratio.

Unlike discrete gearboxes, a CVT provides a continuously varying ratio. Torque transfer is governed by belt traction, normal clamping forces, and frictional interaction between the belt and sheave faces.

When transmitted torque exceeds traction capacity, slip occurs. Accurately modeling both stick and slip regimes is essential for predicting performance.

1.5 Modeling Approach

The system is modeled using a minimal set of dynamic state variables, defined in [Section 7](#). All other quantities are derived algebraically from these states.

The derivation proceeds as follows:

- Governing physical laws are defined ([Section 6](#)),
- Kinematic relationships between pulley geometry and ratio are derived,
- Axial and radial force models are constructed,
- Belt traction limits are formulated,
- Rotational and axial equations of motion are assembled,
- Slip behavior is incorporated.

The final result is a coupled nonlinear dynamic system suitable for time-domain simulation.

2 Notation and Conventions

2.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

2.1.1 Vehicle Longitudinal Direction

The vehicle is modeled as a one-dimensional system moving along a single longitudinal axis.

- Positive direction ($+x$) is defined as forward vehicle motion.
- Vehicle velocity $v > 0$ corresponds to forward motion.
- Resistive forces (rolling resistance, aerodynamic drag, grade load) act in the negative ($-x$) direction.

Incline Angle If terrain grade is considered, the incline angle α is defined as:

- $\alpha > 0$ for uphill motion.
- $\alpha = 0$ for level ground.

The gravitational component acting along the longitudinal axis is:

$$F_{\text{grade}} = mg \sin \alpha$$

This modifies the longitudinal load only and does not introduce additional spatial dimensions into the model.

2.1.2 Rotational Sign Convention

All angular velocities follow the same convention:

- Positive angular velocity corresponds to forward vehicle motion.
- Primary and secondary pulleys are defined such that $\omega_p > 0$ and $\omega_s > 0$ produce forward belt motion.
- Applied engine torque is positive when it increases ω_p .

Reaction torques opposing motion are therefore negative.

2.1.3 CVT Axial Direction

The axial direction of the primary pulley is defined along its shaft axis.

- Positive axial displacement $s > 0$ corresponds to increasing primary effective radius.
- Increasing s therefore corresponds to an upshift (decreasing overall CVT ratio).

The axial velocity $\dot{s}$ follows the same sign convention.

2.1.4 Radial Direction

Radial direction is defined perpendicular to the pulley axis, measured outward from the center of rotation.

- Radial forces $F_r > 0$ act outward from the pulley center.
- Belt normal forces are treated as positive when compressive.

2.1.5 Slip Velocity Sign Convention

Slip velocity between belt and pulley is defined as:

$$v_{\text{slip}} = r_p(s)\omega_p - r_s(s)\omega_s$$

- $v_{\text{slip}} > 0$ indicates the primary surface speed exceeds the secondary surface speed.
- $v_{\text{slip}} = 0$ corresponds to no-slip contact.

2.2 Symbol Summary

Symbol	Meaning
v	vehicle speed
ω_e	engine angular speed
s	primary axial shift position
$\dot{s}$	primary axial shift velocity
R	CVT ratio (define direction explicitly)

Table 1: Core symbols used throughout the derivation.

3 Units and Dimensions

All quantities in this document are expressed in SI units unless otherwise stated.

3.1 Base Units

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 2: SI base units used throughout the model.

3.2 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s ²
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	rad/s ²
Force	newton	N
Torque	newton meter	N m
Moment of inertia	kilogram meter squared	kg m ²
Power	watt	W

Table 3: Common derived SI units used in the CVT model.

4 Constants and Parameters

All physical constants, geometric parameters, and calibrated values used in the model are listed below.

Symbol	Name	Value	Unit	Description
m	Vehicle mass	–	kg	Total vehicle mass
r_w	Wheel radius	–	m	Effective rolling radius
G	Final drive ratio	–	–	Secondary-to-wheel speed ratio
I_p	Primary inertia	–	kg m ²	Lumped primary-side inertia
I_s	Secondary inertia	–	kg m ²	Lumped secondary-side inertia
k_p	Primary spring constant	–	N/m	Primary compression spring
k_s	Secondary spring constant	–	N/m	Secondary compression spring
μ	Friction coefficient	–	–	Belt-pulley friction coefficient
β	Wrap angle	–	rad	Belt wrap angle

Table 4: Model constants and parameters.

5 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

5.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

5.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

5.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Permanent creep and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

5.4 Dynamic and Inertial Modeling

Assumption 9 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 10 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 11 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 12 (Ideal CVT Power Transmission)

Power transmission through the CVT is modeled as ideal (i.e., no internal dissipation). Although belt slip is explicitly modeled and affects torque capacity and shift dynamics, the associated power loss P_{loss} is neglected.

5.5 Environmental and Operational Scope

Assumption 13 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 14 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

6 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Classical mechanics (textbook / lecture notes).

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m^2)
- α is angular acceleration (rad/s^2)

Source

Classical rigid body dynamics (textbook / lecture notes).

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v .

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Bogna Szyk and Steven Wooding (2024).

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

[Encyclopaedia Britannica \(2024\)](#).

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

[Dr. Templeman \(2024\)](#).

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

2024 Power Test, LLC (2024).

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Gearing (Torque Scaling)

Equation

$$\tau_{\text{out}} = \tau_{\text{in}} R_{\text{gear}}$$

Description

- τ_{out} is output torque (N m)
- τ_{in} is input torque (N m)
- R_{gear} is gear reduction ratio (unitless)

Source

Brain (2023).

Derivation

Ideal gear relationship assuming no losses; power balance implies inverse scaling of speed.

Theory 10: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

[Wikipedia contributors](#) (2024).

Derivation

Coulomb friction model for impending slip.

Theory 11: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

[Hackaday](#) (2021).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{dT}{T} = \mu \, d\theta$, then integrate over wrap.

Theory 12: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m³)
- m is mass (kg)
- V is volume (m³)

Source

[The Editors of Encyclopaedia Britannica \(2024\)](#).

Derivation

Definition of density.

Theory 13: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m³)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m²)

Source

[SoftSchools \(2020\)](#).

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 14: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

?.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 15: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

?.

Derivation

Near Earth's surface, $g \approx 9.81 \text{ m/s}^2$. On an incline with angle θ , the component parallel to the surface is $F_g \sin \theta$.

7 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future value depends on its current value and its time derivative. In numerical simulation, state variables are the quantities that are directly integrated forward in time at each timestep.

All other quantities in the model (e.g., torque transfer, CVT ratio, belt forces) are computed algebraically from these states.

7.1 State Vector

State Vector	(7.1)
$\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \end{bmatrix}$	

where:

- ω_p is the angular speed of the primary pulley (rad/s).
- ω_s is the angular speed of the secondary pulley (rad/s).
- s is the axial shift position of the primary sheave (m).
- $\dot{s}$ is the axial shift velocity (m/s).

7.2 Why These Are the State Variables

These four variables represent the independent dynamic degrees of freedom of the CVT system:

- The primary pulley rotation (ω_p) evolves according to applied engine torque and belt reaction torque.
- The secondary pulley rotation (ω_s) evolves according to belt torque and external load torque.
- The axial shift position (s) evolves according to the balance of axial forces between the primary and secondary assemblies.
- The axial velocity ($\dot{s}$) is required because axial motion follows a second-order differential equation.

This will be made clear in the following derivations in later sections (TODO: MAKE BETTER). Removing any of these variables would eliminate an independent dynamic degree of freedom from the model.

7.3 Numerical Integration Perspective

At each simulation timestep Δt , the following updates occur:

- ω_p is updated using its angular acceleration.
- ω_s is updated using its angular acceleration.
- s is updated using $\dot{s}$.
- $\dot{s}$ is updated using axial acceleration.

Thus, the state vector evolves according to differential equations derived in later sections.

All other model quantities — including CVT ratio, belt slip velocity, radial forces, and torque capacity — are computed directly from the current values of ω_p , ω_s , and s .

8 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies effective pulley radii and therefore transmission ratio. The derivations presented here are purely kinematic and do not involve force balances.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

8.1 Conceptual Overview of Variable Geometry

Each pulley in the CVT consists of two opposing sheaves. One sheave is fixed axially, while the other is free to translate along the shaft.

The effective belt radius is determined by the axial separation between these sheaves. As the moveable sheave approaches the fixed sheave, the belt is forced outward along the conical faces, increasing the effective pulley radius.

Conversely, as the sheaves separate, the belt rides deeper between the sheaves, decreasing the effective radius.

Because the belt length is constant, any increase in the primary pulley radius must be accompanied by a corresponding decrease in the secondary pulley radius. This geometric coupling is fundamental to the operation of the CVT and is derived explicitly in the following sections.

TODO: ADD ILLUSTRATION. Cvt cross section of single pulley in 0 shift and full shift

8.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the moveable primary sheave, defined according to the sign convention in [Section 2.1](#).

- At $s = 0$, the primary sheaves are at their maximum axial separation.
- At $s = 0$, the primary pulley is at its minimum effective radius.

For $s > 0$, the moveable sheave translates toward the fixed sheave, reducing the axial gap between the sheaves.

Deadzone and Travel Limits The shift coordinate is bounded as

$$0 \leq s \leq s_{\max}.$$

A deadzone region exists such that

$$0 \leq s < s_{dz}$$

does not alter the belt contact radius on the sheaves.

For

$$s_{dz} \leq s \leq s_{\max},$$

axial motion produces a proportional change in primary radius.

Here:

- s_{dz} denotes the deadzone displacement,
- $s_{\max}$ denotes the maximum allowable sheave travel.

8.3 Primary Outer Radius as a Function of Shift

Given sheave half-angle β , axial closure beyond the deadzone produces a radial increase governed by simple trigonometry.

For an incremental axial displacement Δs beyond s_{dz} , the corresponding radial increase is

$$\Delta r = \frac{\Delta s}{2 \tan \beta}.$$

The factor of 2 arises because axial closure is shared symmetrically between the two opposing sheave faces. The geometric relation is illustrated in [Figure 1](#).

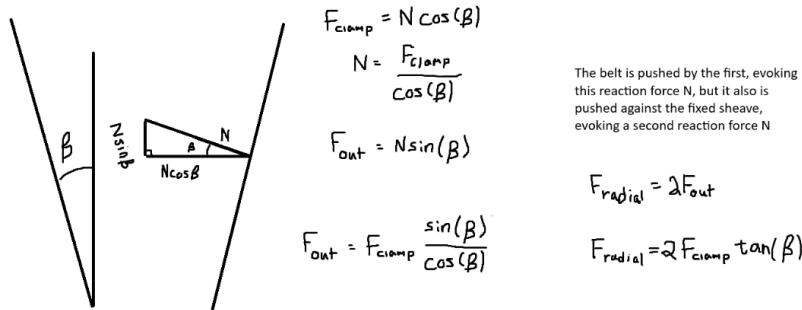


Figure 1: Axial-radial relationship for opposing conical sheaves with half-angle β .

Piecewise Radius Model Let $r_{p,\min}$ denote the minimum primary outer radius. Then the primary outer radius is

$$\text{Primary Outer Radius} \tag{8.1}$$

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\min}, & 0 \leq s < s_{dz}, \\ r_{p,\min} + \frac{s - s_{dz}}{2 \tan \beta}, & s_{dz} \leq s \leq s_{\max}. \end{cases}$$

Remark on Belt Height and Effective Torque Radius

The belt possesses a finite radial thickness h_b . In this section, all geometric relations are formulated using the *outer surface* of the belt as the reference radius. That is, r_{outer} denotes the radial position of the belt's outer face.

This convention simplifies the geometric derivations of pulley radius and belt length. However, torque transmission does not occur at the outer surface. The effective moment arm corresponds approximately to the mid-height of the belt.

Accordingly, the effective torque radius used in force and torque calculations will later be defined as

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}. \quad (8.2)$$

This correction is applied explicitly in the torque transfer formulation (see [Section 8.7](#)).

8.4 Belt Length Constraint

This section derives the geometric constraint relating belt length to the primary radius r_p , secondary radius r_s , and center-to-center distance C . The derivation follows directly from the belt path geometry shown in [Figure 2](#).

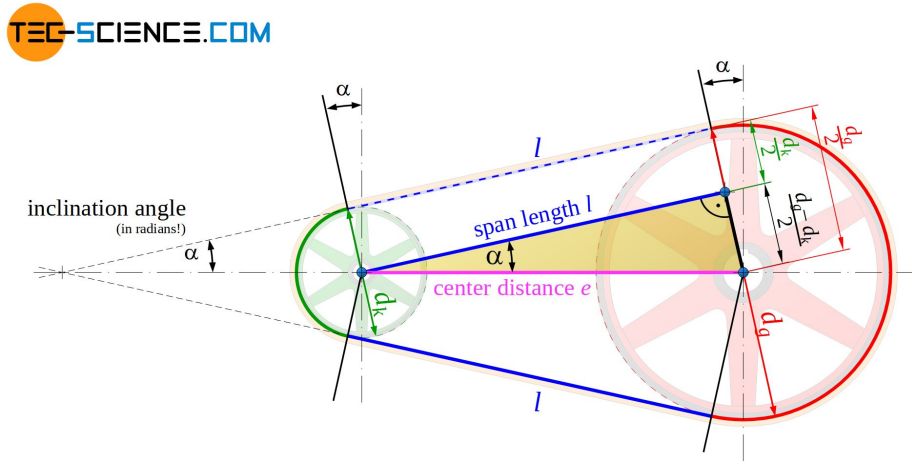


Figure 2: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C . The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans.

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p \phi_p + r_s \phi_s + 2\ell, \quad (8.3)$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length Let ℓ denote the length of the straight belt segment between tangent points. From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \quad (8.4)$$

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in [Figure 2](#)). We define

$$\sin \lambda = \frac{r_s - r_p}{C}, \quad (8.5)$$

and equivalently,

$$\lambda = \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \quad (8.6)$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \quad (8.7)$$

$$\phi_s = \pi + 2\lambda. \quad (8.8)$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Closed Form Belt-Length Expression Substituting [Equation \(8.4\)](#) and [Equation \(8.7\)](#)–[Equation \(8.8\)](#) into [Equation \(8.3\)](#) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \quad (8.9)$$

Finally, substituting [Equation \(8.6\)](#) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (8.10)$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (8.10) yields

$$\text{Belt Length Constraint Root Function} \quad (8.11)$$

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b.$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \quad (8.12)$$

Remark (Transcendental Nature and Approximation)

Equation (8.12) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.

In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.

It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.

However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix ?? quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

8.5 Center-to-Center Distance Determination

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C , we evaluate the belt-length constraint Equation (8.12) at the known initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\min}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\max}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (8.12) yields

$$f(r_{p,\min}, r_{s,\max}, C) = 0. \quad (8.13)$$

This equation contains a single unknown, C , and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b .

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

8.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 8.5, and the primary radius $r_p(s)$ defined by Equation (8.1), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s , the primary radius $r_p(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_p(s), r_s, C) = 0, \quad (8.14)$$

where $f(\cdot)$ is defined in Equation (8.11).

Numerical Determination of $r_s(s)$ Because Equation (8.14) is transcendental in r_s , the secondary radius is obtained numerically for each value of s :

Secondary Outer Radius

(8.15)

$$r_{s,\text{outer}}(s) = \underset{r_s}{\text{root}} \ f(r_p(s), r_s, C).$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in Section 8.5, which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

Wrap Angles as Explicit Functions of Shift Using the geometric relations from [Section 8.4](#), the tangency angle evaluated at the current shift is

$$\lambda(s) = \sin^{-1} \left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C} \right). \quad (8.16)$$

Substituting directly into the wrap-angle relations [Equation \(8.7\)](#) and [Equation \(8.8\)](#) yields the fully expanded expressions.

$$\phi_p(s) = \pi - 2 \sin^{-1} \left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C} \right), \quad (8.17)$$

$$\phi_s(s) = \pi + 2 \sin^{-1} \left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C} \right). \quad (8.18)$$

8.7 Effective Torque Radius

As noted previously, the belt possesses a finite radial thickness h_b . All geometric derivations were performed using the outer belt surface for convenience. However, torque is transmitted through the belt cross-section, and the effective moment arm corresponds approximately to the mid-height of the belt.

The effective torque radius for each pulley is therefore defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \quad (8.19)$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \quad (8.20)$$

These effective radii are used in all subsequent torque transmission calculations.

8.8 CVT Ratio Definition

The instantaneous CVT ratio is defined as the ratio of secondary to primary effective radii:

CVT Ratio	(8.21)
$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}.$	

Because $r_{p,\text{eff}}(s)$ increases with shift s , the ratio $R(s)$ decreases as the primary closes.

Thus:

- Small s (minimum primary radius) corresponds to high ratio.
- Large s (maximum primary radius) corresponds to low ratio.

The transmission ratio is therefore a purely geometric function of shift position.

8.9 Ratio Rate of Change

During shifting, the ratio R varies because the effective pulley radii vary with the shift coordinate s . This section derives an expression for $\dot{R}$ in terms of $\dot{s}$ and geometric derivatives.

8.9.1 Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \quad (8.22)$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds} \dot{s}. \quad (8.23)$$

Thus, the objective is to compute dR/ds .

8.9.2 Differentiate R with Respect to s

Differentiating Equation (8.22) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds} \left(\frac{r_{s,\text{eff}}}{r_{p,\text{eff}}} \right) = \frac{\left(\frac{dr_{s,\text{eff}}}{ds} \right) r_{p,\text{eff}} - r_{s,\text{eff}} \left(\frac{dr_{p,\text{eff}}}{ds} \right)}{r_{p,\text{eff}}^2}. \quad (8.24)$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}} \frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \quad (8.25)$$

At this point, the remaining task is to express the two derivatives $dr_{p,\text{eff}}/ds$ and $dr_{s,\text{eff}}/ds$ in computable form.

8.9.3 Eliminating dr_s/ds Using the Belt-Length Constraint

The secondary radius $r_s(s)$ is defined implicitly through the belt-length constraint and does not admit a closed-form expression in terms of r_p . Direct differentiation of $r_s(s)$ is therefore not practical.

Instead, we differentiate the belt-length compatibility constraint derived in Section 8.4, which is written as

$$f(r_p, r_s, C) = 0, \quad (8.26)$$

with $f(\cdot)$ defined explicitly in Equation (8.11) and C constant. Since $r_p = r_p(s)$ and $r_s = r_s(s)$, differentiating Equation (8.26) with respect to s gives (multivariable chain rule)

$$\frac{\partial f}{\partial r_p} \frac{dr_p}{ds} + \frac{\partial f}{\partial r_s} \frac{dr_s}{ds} = 0. \quad (8.27)$$

Solving Equation (8.27) for dr_s/ds yields

$$\frac{dr_s}{ds} = - \frac{\frac{\partial f}{\partial r_p} \frac{dr_p}{ds}}{\frac{\partial f}{\partial r_s}}. \quad (8.28)$$

8.9.4 Single Expression for dR/ds in Terms of Remaining Unknowns

Substituting Equation (8.28) into Equation (8.25) eliminates dr_s/ds and gives a single expression for dR/ds in terms of the belt constraint partial derivatives and the primary geometric derivative:

$$\frac{dR}{ds} = - \frac{1}{r_{p,\text{eff}}} \frac{\frac{\partial f}{\partial r_p} \frac{dr_p}{ds}}{\frac{\partial f}{\partial r_s}} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \quad (8.29)$$

Equation (8.29) makes the remaining requirements explicit. To evaluate dR/ds (and therefore $\dot{R}$ via Equation (8.23)), three quantities are required:

1. $\frac{\partial f}{\partial r_p}$,
2. $\frac{\partial f}{\partial r_s}$,
3. $\frac{dr_p}{ds}$ (and the associated $\frac{dr_{p,\text{eff}}}{ds}$).

The following subsections derive these three quantities for the specific belt-length function f defined in Equation (8.11) and for the primary geometry defined in Equation (8.1).

8.9.5 Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \quad (8.30)$$

We now compute the two partial derivatives required in Equation (8.28), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p} \left[(r_s - r_p) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) \right] = \underbrace{\frac{\partial}{\partial r_p} (r_s - r_p)}_{(1)} \sin^{-1} \left(\frac{r_s - r_p}{C} \right) + (r_s - r_p) \underbrace{\frac{\partial}{\partial r_p} \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}_{(2)}. \quad (8.31)$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p} (r_s - r_p) = -1. \quad (8.32)$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p} \sin^{-1} \left(\frac{r_s - r_p}{C} \right) = \frac{1}{\sqrt{1-u^2}} \left(-\frac{1}{C} \right). \quad (8.33)$$

Now simplify the denominator:

$$\sqrt{1-u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C} \right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into [Equation \(8.33\)](#) yields

$$\frac{\partial}{\partial r_p} \sin^{-1} \left(\frac{r_s - r_p}{C} \right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \quad (8.34)$$

Substitute into the product rule Substituting [Equation \(8.32\)](#) and [Equation \(8.34\)](#) into [Equation \(8.31\)](#) gives

$$\begin{aligned} \frac{\partial}{\partial r_p} \left[(r_s - r_p) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) \right] &= (-1) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) + (r_s - r_p) \left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} \right) \\ &= -\sin^{-1} \left(\frac{r_s - r_p}{C} \right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \quad (8.35)$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p} \left[2\sqrt{C^2 - (r_s - r_p)^2} \right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \quad (8.36)$$

Assemble $\partial f/\partial r_p$ Differentiating Equation (8.30) term-by-term gives

$$\begin{aligned} \frac{\partial f}{\partial r_p} &= \pi + 2 \left(-\sin^{-1} \left(\frac{r_s - r_p}{C} \right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}} \right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \end{aligned} \quad (8.37)$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s} \left[(r_s - r_p) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) \right] = \sin^{-1} \left(\frac{r_s - r_p}{C} \right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}, \quad (8.38)$$

and

$$\frac{\partial}{\partial r_s} \left[2\sqrt{C^2 - (r_s - r_p)^2} \right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \quad (8.39)$$

Assembling terms gives

$$\begin{aligned} \frac{\partial f}{\partial r_s} &= \pi + 2 \left(\sin^{-1} \left(\frac{r_s - r_p}{C} \right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}} \right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \end{aligned} \quad (8.40)$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (8.30) are

$$\frac{\partial f}{\partial r_p} = \pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right), \quad (8.41)$$

$$\frac{\partial f}{\partial r_s} = \pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \quad (8.42)$$

These expressions are globally valid under the geometric definition of λ given in Equation (8.6) and require no piecewise case distinction.

8.9.6 Primary Radius Derivative

From the piecewise primary radius model Equation (8.1), the primary outer radius satisfies

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{min}}, & 0 \leq s < s_{\text{dz}}, \\ r_{p,\text{min}} + \frac{s - s_{\text{dz}}}{2 \tan \beta}, & s_{\text{dz}} \leq s \leq s_{\text{max}}. \end{cases}$$

Differentiating with respect to s gives

$$\frac{dr_p}{ds} = \frac{dr_{p,\text{outer}}}{ds} = \begin{cases} 0, & 0 \leq s < s_{\text{dz}}, \\ \frac{1}{2 \tan \beta}, & s_{\text{dz}} \leq s \leq s_{\text{max}}. \end{cases} \quad (8.43)$$

Because $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\text{eff}}}{ds} = \frac{dr_p}{ds}. \quad (8.44)$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit s_{max} , then $r_{p,\text{outer}}$ (and thus $r_{p,\text{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\text{max}}$. In this document we restrict to the admissible domain $0 \leq s \leq s_{\text{max}}$.

8.9.7 Final Expression for dR/ds and $\dot{R}$

Equation (8.29) expresses dR/ds in terms of three quantities: (i) the belt-constraint partial derivatives $\partial f/\partial r_p$ and $\partial f/\partial r_s$, and (ii) the primary geometric derivatives dr_p/ds and $dr_{p,\text{eff}}/ds$.

From Section 8.9.5, the partial derivatives of the belt-length constraint Equation (8.11) are

$$\frac{\partial f}{\partial r_p} = \pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right), \quad (8.45)$$

$$\frac{\partial f}{\partial r_s} = \pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \quad (8.46)$$

From Section 8.9.6, the primary outer-radius derivative is Equation (8.43), and since $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ with h_b constant, Equation (8.44) gives $dr_{p,\text{eff}}/ds = dr_p/ds$.

Substituting Equation (8.45)–Equation (8.46) and Equation (8.44) into Equation (8.29) yields

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}} \frac{\pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}{\pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right] \frac{dr_p}{ds}. \quad (8.47)$$

Finally, substituting the piecewise primary derivative Equation (8.43) gives the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ \frac{1}{2 \tan \beta} \left[-\frac{1}{r_{p,\text{eff}}} \frac{\pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}{\pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right], & s_{dz} \leq s \leq s_{\max}. \end{cases} \quad (8.48)$$

Using the chain rule Equation (8.23), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (8.48) and rearranging gives

Rate of Change of Transmission Ratio	(8.49)
$\dot{R} = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\dot{s}}{2 \tan \beta r_{p,\text{eff}}} \left[\frac{\pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}{\pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)} + \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}} \right], & s_{dz} \leq s \leq s_{\max}. \end{cases}$	

Summary Equations Equation (8.48) and Equation (8.49) provide a complete closed-form expression for the geometric sensitivity of the transmission ratio and its time rate of change.

Specifically,

- $\frac{dR}{ds}$ quantifies how sensitive the CVT ratio is to axial sheave motion.
- $\dot{R} = \frac{dR}{dt}$ relates axial shift velocity $\dot{s}$ directly to rotational ratio dynamics.

The result captures three coupled geometric effects:

1. The primary sheave cone geometry through β ,
2. The nonlinear belt-length compatibility through the inverse-sine term,
3. The redistribution of effective radii between pulleys during shift.

The deadzone behavior is explicitly represented: when $0 \leq s < s_{dz}$, the effective radii are constant and $\dot{R} = 0$.

In the active shift region $s_{dz} \leq s \leq s_{\max}$, ratio dynamics are governed entirely by geometric coupling and axial shift velocity.

This completes the purely kinematic derivation of CVT ratio evolution. Subsequent sections incorporate force balance and torque transfer to determine $\dot{s}$ dynamically.

8.10 Section Summary and Forward Connection

This section has established a complete *geometric foundation* for the CVT model. The development was intentionally kinematic: no force balances, no slip assumptions, and no torque relations were introduced. The objective was to construct a mathematically closed mapping

$$s(t) \longrightarrow r_{p,\text{eff}}(s), r_{s,\text{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with all intermediate relationships derived explicitly.

What Has Been Established The following key geometric components are now rigorously defined:

1. **Primary Radius Law** Equation (8.1) provides the piecewise relationship between axial shift s and primary outer radius, governed by the cone half-angle β and deadzone s_{dz} .
2. **Belt-Length Compatibility Constraint** Equation (8.12) defines the transcendental geometric coupling between r_p , r_s , and center distance C . This constraint enforces constant belt length and represents the core geometric coupling of the transmission.
3. **Center Distance Determination** Section 8.5 determines C once during initialization by solving Equation (8.13). After this step, C is fixed for the remainder of the model.
4. **Secondary Radius Implicit Law** Equation (8.14) and Equation (8.15) define $r_s(s)$ implicitly via the belt constraint, producing the geometric redistribution of radius during shift.
5. **Effective Torque Radii** Section 8.7 corrects outer radii to effective radii via Equation (8.2), ensuring torque calculations later act at the belt mid-height.
6. **Transmission Ratio** Equation (8.21) defines the purely geometric ratio

$$R(s) = \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}.$$

7. **Ratio Sensitivity and Time Evolution** Section 8.9 culminates in the closed-form expressions Equation (8.48) and Equation (8.49), which provide

$$\frac{dR}{ds} \quad \text{and} \quad \dot{R} = \frac{dR}{ds} \dot{s}.$$

These expressions fully capture the nonlinear geometric sensitivity introduced by belt-length compatibility and cone geometry.

Structural Importance for the Full Model This section establishes the geometric backbone of the entire CVT simulation framework. Every dynamic result that follows will rely on the mappings derived here.

In particular:

- Torque transfer equations will use $r_{p,\text{eff}}$ and $r_{s,\text{eff}}$ from Section 8.7.
- Axial force balances will determine $\dot{s}$ dynamically.

- The ratio dynamics in Equation (8.49) will convert axial motion into rotational acceleration coupling between shafts.
- Slip models and capstan friction relations will depend explicitly on wrap angles Equation (8.7)–Equation (8.8) and the geometric redistribution captured in Equation (8.11).
- Power flow and coupling torque formulations will use $R(s)$ and $\dot{R}$ directly to relate engine dynamics to driven-shaft dynamics.

Thus, this section isolates geometry from physics. All subsequent force and torque derivations will treat the geometric relationships derived here as fixed, differentiable mappings.

Key Modeling Properties Several important structural properties are now clear:

1. The ratio evolution is inherently nonlinear due to the inverse-sine term in the belt constraint.
2. The geometric coupling is globally valid (no small-angle approximation required).
3. The deadzone is explicitly represented: for $0 \leq s < s_{dz}$, R is constant.
4. Ratio sensitivity depends on both cone angle β and instantaneous pulley radii.
5. The system is fully differentiable within the admissible domain $s_{dz} \leq s \leq s_{max}$, enabling continuous-time simulation and ODE integration.

Transition to Dynamics With geometry now fully specified, the model is ready to incorporate force balance and torque transfer.

The next major objective is to determine $\dot{s}$ from axial forces (centrifugal force, spring force, belt tension, and normal reaction forces). Once $\dot{s}$ is known, the ratio evolution Equation (8.49) immediately provides rotational coupling between shafts.

In other words:

$$\text{Axial Dynamics} \longrightarrow \dot{s} \longrightarrow \dot{R} \longrightarrow \text{Rotational Dynamics}.$$

This completes the geometric phase of the CVT derivation. The model is now prepared for the introduction of force and torque equilibrium in the subsequent section.

9 Pulley Axial Force Models

This section introduces the axial force models that drive sheave motion and therefore determine the evolution of the shift coordinate $s(t)$.

The state vector $\mathbf{x}(t)$ was defined in Section 7 (see Equation (7.1)) as $(\omega_p, \omega_s, s, \dot{s})$. The objective of the present section is to derive expressions for the axial forces acting on the primary and secondary sheaves, which will ultimately provide the equation of motion for $\ddot{s}$.

Throughout this section, axial and radial directions follow the conventions established in Section 2.1: positive axial displacement corresponds to increasing primary effective radius, and radial forces are positive outward from the pulley center.

The goal is to express the net axial forces generated by the primary and secondary pulley mechanisms as functions of the current state. These forces will later be combined using **Theory 1 (Newton's Second Law (Translation))** to obtain an equation of motion for s (and therefore $\dot{s}$ and $\ddot{s}$).

Although the specific models developed here correspond to the hardware considered in this work (flyweight/ramp actuation on the primary and a torque-reactive helix with springs on the secondary), the formulation is modular: alternative actuator models may be substituted provided they return the correct net axial force.

9.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined surfaces. To keep the force models general and drop-in replaceable, we represent each mechanism's geometry using user-supplied functions of the axial shift coordinate s .

Primary Ramp Geometry (Axial-to-Radial Profile) On the primary pulley, centrifugal (radial) force is redirected into axial closing force through a ramp. We parameterize the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{9.1}$$

where $r_f(s)$ is the effective flyweight radius.

Because s is axial distance, the local ramp slope is dr_f/ds . The corresponding ramp angle is defined consistently as

$$\alpha_p(s) \equiv \tan^{-1}\left(\frac{dr_f}{ds}\right), \quad \text{so that} \quad \tan \alpha_p(s) = \frac{dr_f}{ds}. \tag{9.2}$$

This definition is physically intuitive: a steeper ramp yields a larger change in radius per unit axial travel and therefore larger axial gain through the $\tan \alpha_p$ factor used later in the force model.

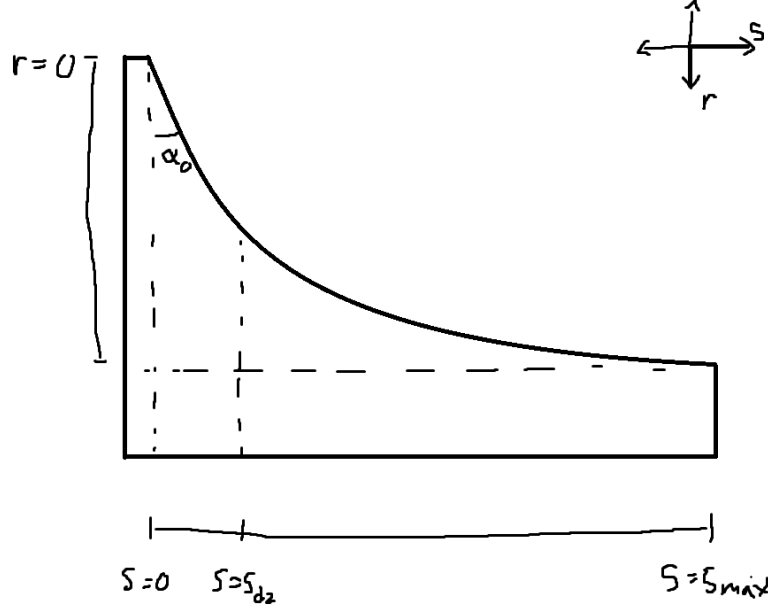


Figure 3: Primary ramp geometry. The flyweight radius is parameterized by $r_f(s)$, where s is axial travel. The local ramp angle is defined by $\tan \alpha_p(s) = dr_f/ds$, so steeper profiles (larger dr_f/ds) redirect a larger fraction of radial force into axial closing force.

Primary Admissibility Conditions The primary ramp profile $r_f(s)$ must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\frac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\frac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan \alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces a relative rotation of the helix mechanism, loading the torsional spring and generating torque-reactive clamping. We parameterize this kinematically as

$$\theta_s = \theta_s(s), \quad (9.3)$$

where $\theta_s(s)$ is the helix-induced rotation.

The quantity $d\theta_s/ds$ is the helix “lead” (rotation per unit axial travel). Under the convention used in this work, *shallower* helix ramps produce *more* rotation for the same axial travel, while *steeper* ramps produce *less* rotation. To encode this consistently, we define the effective helix angle $\alpha_s(s)$

by the inverse-slope relationship

$$\tan \alpha_s(s) \equiv \frac{1}{r_h \frac{d\theta_s}{ds}}, \quad (9.4)$$

where r_h is the effective helix radius. With this definition, a decrease in α_s (a “shallower” ramp) corresponds to an increase in $d\theta_s/ds$, matching the physical behavior described above.

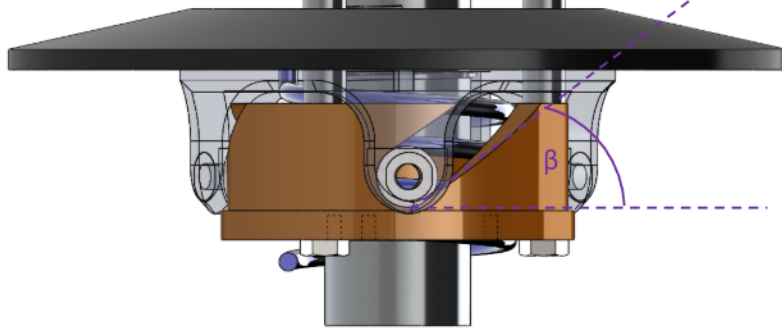


Figure 4: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ of the helix mechanism. The helix angle is defined so that shallower ramps produce greater rotation per unit axial travel: $\tan \alpha_s(s) = (r_h d\theta_s/ds)^{-1}$. Diagram should be 3d, show clearly which direction shift distance is in (and why the angle is defined differently). Use dotted lines to show a single angle along the ramp (instantaneous slope or smt)

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\frac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\frac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases torsional loading. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix ??.

9.2 Primary Axial Force Model

The primary axial clamping force arises from centrifugal loading of the flyweights and opposition from the primary compression spring.

Figure 5: Primary CVT assembly showing flyweight mass m_f , ramp profile $r_f(s)$, compression spring with stiffness k_p and preload $x_{p,0}$, and axial shift coordinate s . TODO: use 3D model to define constants and show various aspects of primary CVT.

The state variable governing primary rotation is ω_p .

Centrifugal Loading of Flyweights From [Theory 5 \(Centrifugal Force\)](#),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_f(s)$ denote the effective flyweight radius.

The radial centrifugal force is therefore

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 r_f(s). \quad (9.5)$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in [Figure 6](#), the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \quad (9.6)$$

From [Equation \(9.2\)](#), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting this into Equation (9.6) gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 r_f(s) \frac{dr_f}{ds}. \quad (9.7)$$

Thus the axial gain of centrifugal force is directly governed by the local slope of the ramp profile.

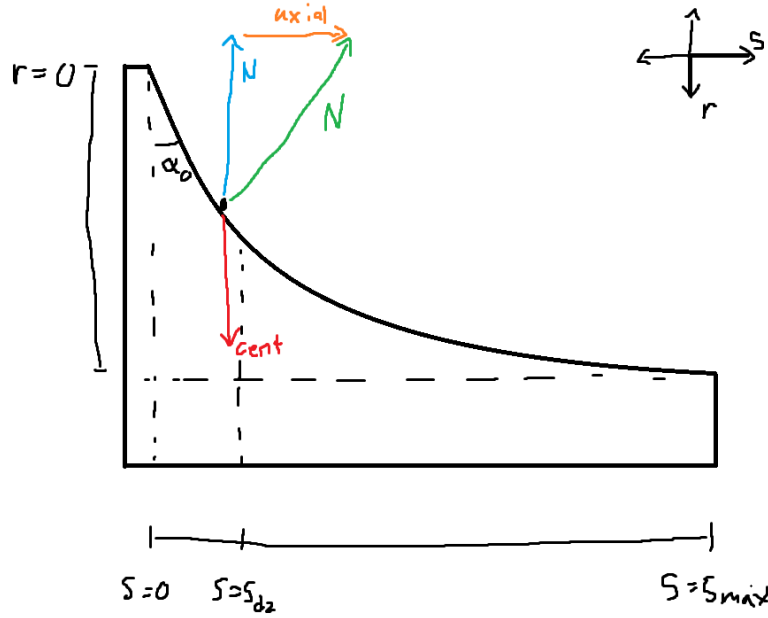


Figure 6: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \quad (9.8)$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

$$F_{p,ax}(s, \omega_p) = m_f \omega_p^2 r_f(s) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$$

9.3 Secondary Axial Force Model

The secondary pulley is *torque-reactive*: its axial clamping force is generated in response to torque transmitted through the CVT, rather than from rotational speed (as on the primary). This fundamental difference determines the load-sensitive behavior of the secondary mechanism.

Figure 7: Secondary CVT assembly showing the helix mechanism, torsional spring with stiffness $k_{s,\theta}$ and preload $\theta_{s,0}$, compression spring with stiffness $k_{s,x}$ and preload $x_{s,0}$, and effective helix radius r_h . The torque τ_s transmitted through the belt creates a tangential force at the helix, which is redirected into axial clamping force through the inclined helix ramps. TODO: use 3D model to show helix geometry, torque path, and direction of resulting axial force.

Why the Secondary Is Torque-Reactive On the primary pulley, centrifugal force from the spinning flyweights generates axial clamping—this depends on rotational *speed* (ω_p), not on the torque being transmitted. The primary therefore tends to upshift at high RPM regardless of load.

The secondary operates on a fundamentally different principle: it generates axial clamping force in response to torque *transmitted through the CVT*. To understand why, consider the mechanics of force generation at the helix.

The helix mechanism converts tangential force into axial force. However, a tangential force can only exist if there is resistance to rotation. If the secondary pulley were free to spin (i.e., no load on the driven shaft), the belt would simply rotate the pulley without creating any tangential force at the helix contact surface—and therefore no axial clamping would occur.

For a reaction force to exist at the helix, the following must hold:

- The belt must apply a driving torque to the secondary pulley (higher tension on the entry side than the exit side).
- The driven load must resist this torque, preventing free rotation.
- The torque transmitted from belt to load passes through the helix mechanism, creating a tangential force at the helix ramps.
- This tangential force, acting on the inclined helix surface, generates an axial component that clamps the sheaves together.

In other words, the helix “reacts” against the torque being transmitted. The harder the CVT must work (i.e., the more torque flows through the secondary to the load), the harder the secondary clamps the belt.

This distinction is critical:

- The relevant torque is the torque *transmitted through the CVT* to the driven load—not simply the load torque or an externally applied torque.

- If the driven shaft were disconnected, the secondary would spin freely and generate no clamping force, regardless of belt tension.
- Under light load, secondary clamping is weak, allowing the primary to dominate and upshift. Under heavy load, secondary clamping increases, resisting upshift or forcing a downshift.

This torque-reactive behavior provides automatic load sensitivity: the CVT naturally downshifts under high-torque conditions (e.g., hill climbing) and upshifts under light-torque conditions (e.g., cruising).

Definition of Secondary Shaft Torque Let τ_s denote the torque transmitted through the secondary shaft. This is the net torque that flows from the belt, through the secondary pulley mechanism (including the helix), and into the driven load. It is this transmitted torque—not simply the load seen by the shaft—that creates the reaction force at the helix and therefore determines the axial clamping force.

The relationship between τ_s and the belt tension distribution is developed in the following section on belt mechanics; for the present derivation, τ_s is treated as a known input to the secondary axial force model.

Torsional Spring Torque From [Theory 7 \(Hooke's Law \(Torsional\)\)](#),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}(\theta_{s,0} + \theta_s(s)). \quad (9.10)$$

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \quad (9.11)$$

Both torque contributions act in the same rotational direction at the helix: the transmitted torque τ_s loads the helix as it passes from belt to driven shaft, while the torsional spring $\tau_{s,\text{spring}}$ resists the relative rotation induced by axial displacement. Both therefore contribute positively to the tangential force at the helix ramps, and hence to axial clamping. The torsional spring provides a baseline clamping force even when no torque is being transmitted ($\tau_s = 0$), ensuring the belt remains seated at idle.

Torque-to-Axial Conversion Through the Helix From [Theory 3 \(Torque Definition\)](#), $\tau = F_t r_h$, so

$$F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The helix redirects tangential force into axial force. From trigonometry (see [Figure 8](#)),

$$F_{s,\text{helix}} = \frac{F_t}{2 \tan \alpha_s(s)}. \quad (9.12)$$

From [Section 9.1](#), the helix angle satisfies

$$\tan \alpha_s(s) = \frac{1}{r_h \frac{d\theta_s}{ds}}, \quad \Rightarrow \quad \frac{1}{\tan \alpha_s(s)} = r_h \frac{d\theta_s}{ds}.$$

Substituting into [Equation \(9.12\)](#) yields

$$F_{s,\text{helix}}(s, \tau_s) = \frac{\tau_{s,\text{total}}}{2} \frac{d\theta_s}{ds}. \quad (9.13)$$

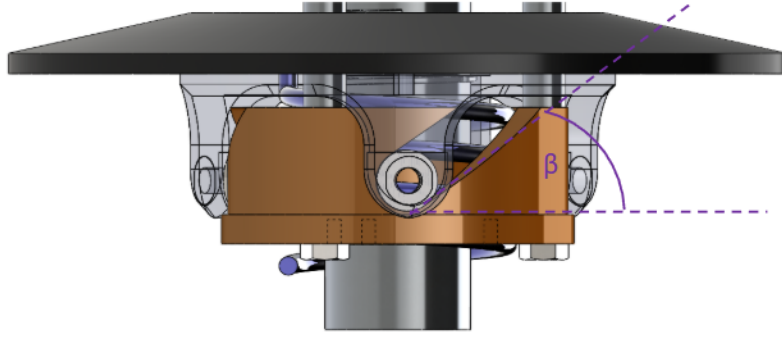


Figure 8: Secondary helix geometry. Tangential force generated by shaft torque is redirected into axial force. The projection introduces a factor $1/\tan \alpha_s$, which equals $r_h d\theta_s/ds$ by definition of the helix mapping. TODO: Add a point with forces (FBD)

Secondary Compression Spring In addition to helix loading, the secondary includes a compression spring. From [Theory 6 \(Hooke's Law \(Compressional\)\)](#),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote the secondary compression stiffness and $x_{s,0}$ preload. With compression increasing with s ,

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \quad (9.14)$$

Net Secondary Axial Force The net secondary axial force is the sum of the helix-generated clamping force and the compression spring force:

$$F_{s,\text{ax}}(s, \tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}(\theta_{s,0} + \theta_s(s))}{2} \frac{d\theta_s}{ds}}_{\text{helix + torsional spring}} + \underbrace{k_{s,x}(x_{s,0} + s)}_{\text{compression spring}}.$$

9.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Centrifugal Force Formulation From [Theory 5 \(Centrifugal Force\)](#), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around a pulley, we require:

1. The mass of belt material in contact with the pulley.
2. The radius at which this mass rotates.
3. The angular velocity of the belt.

Each of these quantities is derived below.

Belt Cross-Section Geometry The belt cross-section is approximated as a trapezoid (see [Figure 9](#)) with:

- radial thickness h_b ,
- outer width b_{out} (at the belt's outer surface),
- inner width b_{in} (at the belt's inner surface).

Figure 9: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b .

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}}). \quad (9.16)$$

Let ρ_b denote the belt material density (kg/m^3). The linear mass density (mass per unit belt length) is then

$$\mu_b = \rho_b A_b. \quad (9.17)$$

Belt Mass Centroid Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner surface. Because the belt cross-section is trapezoidal (not rectangular), the centroid is offset from the geometric center.

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y :

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \quad (9.18)$$

The centroid location Δr_{cm} , measured inward from the outer surface, is computed from the definition of the centroid of an area:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y b(y) \, dy. \quad (9.19)$$

Substituting Equation (9.18) into this integral:

$$\begin{aligned} \int_0^{h_b} y b(y) \, dy &= \int_0^{h_b} y \left[b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right] \, dy \\ &= \int_0^{h_b} b_{\text{out}} y \, dy + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2 \, dy \\ &= b_{\text{out}} \left[\frac{y^2}{2} \right]_0^{h_b} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \left[\frac{y^3}{3} \right]_0^{h_b} \\ &= b_{\text{out}} \cdot \frac{h_b^2}{2} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \cdot \frac{h_b^3}{3} \\ &= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} (b_{\text{in}} - b_{\text{out}}) \\ &= \frac{h_b^2}{6} (3b_{\text{out}} + 2b_{\text{in}} - 2b_{\text{out}}) \\ &= \frac{h_b^2}{6} (b_{\text{out}} + 2b_{\text{in}}). \end{aligned} \quad (9.20)$$

Dividing by the area $A_b = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}})$ from Equation (9.16):

$$\Delta r_{\text{cm}} = \frac{\frac{h_b^2}{6} (b_{\text{out}} + 2b_{\text{in}})}{\frac{h_b}{2} (b_{\text{out}} + b_{\text{in}})} = h_b \cdot \frac{b_{\text{out}} + 2b_{\text{in}}}{3(b_{\text{out}} + b_{\text{in}})}. \quad (9.21)$$

The belt mass-centroid radius at any shift position is therefore

$$r_{\text{cm}}(s) = r_{\text{out}}(s) - \Delta r_{\text{cm}}, \quad (9.22)$$

where $r_{\text{out}}(s)$ is the outer belt radius at the current shift (equal to the pulley outer radius at that position).

Distributed Centrifugal Load Along the Wrap Consider a differential belt element along the wrapped arc. Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped arc spans

$$\theta \in \left[-\frac{\phi}{2}, \frac{\phi}{2} \right],$$

where ϕ is the total wrap angle (see [Figure 10](#)).

Figure 10: Belt wrapped around a pulley. A differential element at angle θ has arc length $d\ell = r_{\text{cm}} d\theta$ and experiences radially outward centrifugal force.

A differential belt element has arc length

$$d\ell = r_{\text{cm}} d\theta,$$

and therefore differential mass

$$dm = \mu_b d\ell = \mu_b r_{\text{cm}} d\theta. \quad (9.23)$$

From [Theory 5 \(Centrifugal Force\)](#), the centrifugal force on this element is

$$dF_c = dm \omega^2 r_{\text{cm}} = \mu_b \omega^2 r_{\text{cm}}^2 d\theta. \quad (9.24)$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating [Equation \(9.24\)](#) over the entire wrap angle yields the total radially-directed centrifugal force:

$$\begin{aligned} F_{c,\text{rad}} &= \int_{-\phi/2}^{\phi/2} \mu_b \omega^2 r_{\text{cm}}^2 d\theta \\ &= \mu_b \omega^2 r_{\text{cm}}^2 \int_{-\phi/2}^{\phi/2} d\theta \\ &= \mu_b \omega^2 r_{\text{cm}}^2 \phi \\ &= \rho_b A_b \omega^2 r_{\text{cm}}^2 \phi. \end{aligned} \quad (9.25)$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β , this radial loading is redirected into an axial component that tends to push the sheaves together (i.e., contributes to clamping force).

From Equation (8.43), the relationship between radial and axial displacement in the active shift region is

$$\frac{dr}{ds} = \frac{1}{2 \tan \beta}.$$

This geometric ratio determines how radial force projects onto the axial direction. Inverting, a radial force F_{rad} produces an axial clamping contribution

$$F_{\text{ax}} = F_{\text{rad}} \cdot \frac{1}{2 \tan \beta}.$$

Substituting the radial centrifugal force from Equation (9.25):

$$F_{c,\text{ax}} = \frac{\rho_b A_b \omega^2 r_{\text{cm}}^2 \phi}{2 \tan \beta}. \quad (9.26)$$

Primary Pulley Contribution On the primary pulley, the belt rotates at approximately $\omega = \omega_p$. The outer belt radius equals the primary outer radius from Equation (8.1), which is piecewise in s and accounts for the deadzone behavior.

The mass-centroid radius is

$$r_{p,\text{cm}}(s) = r_{p,\text{outer}}(s) - \Delta r_{\text{cm}}, \quad (9.27)$$

and the wrap angle $\phi_p(s)$ is given by Equation (8.17).

Substituting into Equation (9.26) yields the primary belt centrifugal contribution to axial clamping:

$$F_{c,p}(s, \omega_p) = \frac{\rho_b A_b \omega_p^2 r_{p,\text{cm}}(s)^2 \phi_p(s)}{2 \tan \beta}. \quad (9.28)$$

Remark (Belt Angular Velocity)

Using the primary angular velocity ω_p in the centrifugal force calculation is an approximation that assumes the belt speed matches the primary sheave speed. In reality, belt slip and compliance may cause the belt to rotate at a vastly different angular velocity than the primary sheave.

Secondary Pulley Contribution On the secondary pulley, the belt rotates at approximately $\omega = \omega_s$. The outer belt radius $r_{s,\text{outer}}(s)$ is determined implicitly by the belt-length constraint (see [Equation \(8.15\)](#)).

The mass-centroid radius is

$$r_{s,\text{cm}}(s) = r_{s,\text{outer}}(s) - \Delta r_{\text{cm}}, \quad (9.29)$$

and the wrap angle $\phi_s(s)$ is given by [Equation \(8.18\)](#).

The secondary belt centrifugal contribution to axial clamping is

$$F_{c,s}(s, \omega_s) = \frac{\rho_b A_b \omega_s^2 r_{s,\text{cm}}(s)^2 \phi_s(s)}{2 \tan \beta}. \quad (9.30)$$

9.5 Axial Shift Dynamics

We now assemble the axial force models into a single equation of motion for the shift coordinate $s(t)$, defined in [Section 7](#) as part of the state vector $(\omega_p, \omega_s, s, \dot{s})$. As established in [Section 8.10](#), the belt-length constraint and conical kinematics fully determine all pulley radii, wrap angles, and span geometry once s is specified, so s serves as a single generalized coordinate for the entire shifting system.

Sign Convention By definition, increasing s corresponds to closing the primary pulley. Because total belt length is fixed, this necessarily induces the complementary motion at the secondary. Under this convention:

- A positive primary axial force $F_{p,\text{ax}}$ tends to increase s .
- A positive secondary axial force $F_{s,\text{ax}}$ tends to decrease s .
- The belt centrifugal forces $F_{c,p}$ and $F_{c,s}$ act in the same direction as their respective pulley's axial force (i.e., $F_{c,p}$ adds to the primary's tendency to increase s , while $F_{c,s}$ adds to the secondary's tendency to decrease s).

This sign structure will be reflected explicitly in the final force balance.

Equivalent Inertia Associated with s Newton's second law applies to physical masses in translation. Since s parameterizes coordinated motion of several bodies, we construct an equivalent mass m_{eq} such that the total kinetic energy associated with shift can be written in the form of [Theory 4 \(Kinetic Energy \(Translational\)\)](#):

$$T_{\text{shift}} = \frac{1}{2} m_{\text{eq}} \dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$ in the positive coordinate direction. The movable secondary sheave translates with equal magnitude speed (but opposite physical direction), so its kinetic energy also contributes proportionally to $\dot{s}^2$.

Let $m_{p,m}$ and $m_{s,m}$ denote the masses of the movable primary and secondary sheaves.

Belt Mass Contribution The total belt mass is

$$m_b = \rho_b A_b L_b, \quad (9.31)$$

where A_b is the trapezoidal cross-sectional area from Equation (9.16) and L_b is the constant belt length.

The belt moves continuously with a single belt speed, but the axial component of that motion associated with a change in s is determined by geometry. From the axial-radial coupling in Equation (8.43), a change in sheave separation s is shared between the two conical faces supporting the belt wedge. Thus, the belt's axial velocity relative to the pulley axis is

$$\dot{s}_b = \frac{\dot{s}}{2}. \quad (9.32)$$

The kinetic energy associated with this axial motion is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2} \right)^2 = \frac{1}{2} \left(\frac{m_b}{4} \right) \dot{s}^2.$$

Expressed in the generalized coordinate s , the belt contributes an effective mass of $m_b/4$.

Total Equivalent Mass The total inertia associated with shift becomes

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \quad (9.33)$$

Net Axial Force in the s Coordinate Collecting the axial force contributions derived previously:

- $F_{p,\text{ax}}(s, \omega_p)$ from Equation (9.9),
- $F_{s,\text{ax}}(s, \tau_s)$ from Equation (9.15),
- $F_{c,p}(s, \omega_p)$ from Equation (9.28),
- $F_{c,s}(s, \omega_s)$ from Equation (9.30).

With the sign convention defined above, the generalized force in the positive s direction is

$$\sum F_s = \left(F_{p,\text{ax}} + F_{c,p} \right) - \left(F_{s,\text{ax}} + F_{c,s} \right). \quad (9.34)$$

Shift Equation of Motion Applying Newton's second law in the generalized coordinate,

$$\sum F_s = m_{\text{eq}} \ddot{s},$$

and substituting Equation (9.33) and Equation (9.34), we obtain

$$\ddot{s} = \frac{\left(F_{p,\text{ax}}(s, \omega_p) + F_{c,p}(s, \omega_p)\right) - \left(F_{s,\text{ax}}(s, \tau_s) + F_{c,s}(s, \omega_s)\right)}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}}.$$

Equation (9.35) provides the axial shift acceleration as a function of the current state variables $(s, \dot{s}, \omega_p, \omega_s)$ and the reacted secondary torque τ_s . The remaining coupling between axial and rotational dynamics enters through τ_s , which will be determined later using [Theory 2 \(Newton's Second Law \(Rotation\)\)](#).

9.6 Section Summary and Forward Outlook

In this section we established a complete axial force model for the CVT mechanism and derived a closed-form equation of motion for the generalized shift coordinate $s(t)$.

Beginning with geometric parameterizations of the primary ramp and secondary helix, we expressed all actuator forces in terms of the state variables $(s, \dot{s}, \omega_p, \omega_s)$. We then incorporated spring forces, torque-reactive loading on the secondary, and the distributed centrifugal loading of the belt. Finally, by constructing an equivalent inertia associated with the coordinated axial motion of the movable sheaves and belt mass, we obtained the governing shift dynamics equation [Equation \(9.35\)](#).

At this stage, the axial subsystem is formally closed *except* for one remaining coupling quantity: the reacted secondary torque τ_s . Although τ_s appears explicitly in the secondary axial force model, its value is determined by rotational dynamics of the drivetrain and by torque transmission through the belt. Aside from the shared state variables, the axial and rotational subsystems are coupled only through this torque term.

The next step is to determine how torque is transmitted across the CVT as a function of pulley radii and speed ratio. This will establish the relationship between τ_s , input torque at the primary, and the evolving ratio $R = r_s/r_p$. Once this transmission relationship is obtained, the axial and rotational equations can be combined into a unified, state-space representation of the full CVT dynamics.

With the axial force structure now complete, attention turns to the torque transmission mechanism that closes the remaining dynamical loop.

10 Torque Transmission and Coupling Torque

The axial shift dynamics derived in [Section 9.5](#) require the reacted secondary torque τ_s appearing in [Equation \(9.15\)](#). In this section we determine the torque transmitted through the CVT as a function of the rotational states and the instantaneous ratio $R(t)$.

We proceed in four steps:

1. Write rotational equations of motion for the primary and secondary pulleys.
2. Define the inertias associated with each pulley.
3. Introduce the CVT ratio kinematics.

4. Eliminate accelerations to obtain the transmitted torque.

The result provides the ideal torque transmission law associated with the geometric ratio $R(t)$. This serves as the baseline torque model and will be generalized in the next section.

10.1 System Definition and Torque Path

Figure 11: Torque transmission schematic (placeholder). Engine torque τ_{eng} drives the primary pulley (angular velocity ω_p). Coupling torque τ_p is transmitted through the belt to the secondary pulley (angular velocity ω_s), where it appears as $\tau_s = R\tau_p$. The secondary delivers torque to the downstream drivetrain, which resists motion with torque τ_{load} .

We model two rotating bodies:

- Primary pulley (rigidly connected to the engine),
- Secondary pulley (driving the downstream drivetrain and vehicle).

Let

- I_p : total rotational inertia of the engine + primary CVT pulley,
- I_s : total rotational inertia about the secondary pulley,
- $\tau_{\text{eng}}(t)$: engine output torque applied to the primary pulley, modeled as a user-supplied function of primary speed (see [Section 10.3](#)),
- $\tau_p(t)$: coupling torque (as seen at the primary pulley),
- $\tau_s(t)$: coupling torque (as seen at the secondary pulley),
- $\tau_{\text{load}}(t)$: resisting torque reflected to the secondary pulley from road loads (see [Section 10.4](#)),

10.2 Rotational Equations of Motion

Applying [Theory 2 \(Newton's Second Law \(Rotation\)\)](#) to each pulley:

Primary Pulley

$$\tau_{\text{eng}}(t) - \tau_p(t) = I_p \alpha_p(t), \quad (10.1)$$

Secondary Pulley

$$\tau_s(t) - \tau_{\text{load}}(t) = I_s \alpha_s(t), \quad (10.2)$$

where

$$\alpha_p(t) = \frac{d\omega_p}{dt}, \quad \alpha_s(t) = \frac{d\omega_s}{dt}.$$

At this stage we have four unknown quantities:

$$\alpha_p(t), \quad \alpha_s(t), \quad \tau_p(t), \quad \tau_s(t),$$

but only two equations ([Equation \(10.1\)](#) and [Equation \(10.2\)](#)). Two additional relations are therefore required to close the system. First, we will define the inertias I_p and I_s in terms of physical

parameters of the system. At this point, the external torques $\tau_{\text{eng}}(t)$ and $\tau_{\text{load}}(t)$ remain to be specified. In the remainder of this section we: (i) define τ_{eng} from the engine torque curve under a full-throttle assumption, and (ii) derive τ_{load} from longitudinal road-load theories and reflect it to the secondary pulley.

10.3 Engine Torque Model

The primary forcing term $\tau_{\text{eng}}(t)$ is supplied by the engine. In this formulation, the engine is modeled as a user-provided torque curve defined as a function of primary angular velocity.

Following [Assumption 14 \(Full-Throttle Engine Model\)](#), we assume full-throttle operation throughout the simulation. Under this assumption,

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}(\omega_p(t)), \quad (10.3)$$

where $\tau_{\text{eng}}(\cdot)$ is a supplied admissible function.

Admissibility Constraints. The engine torque function must satisfy:

- It is defined over a bounded operating domain $\omega_{\min} \leq \omega_p \leq \omega_{\max}$.
- It is finite and real-valued on this domain.
- Outside this domain, torque is either saturated or ω_p is constrained to remain within bounds.

No additional structural assumptions (e.g., polynomial form, closed-form analytic expression) are required by the dynamical model. Only the mapping $\omega_p \mapsto \tau_{\text{eng}}$ is needed to evaluate [Equation \(10.1\)](#).

10.4 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ in [Equation \(10.2\)](#) represents the longitudinal road load reflected to the secondary pulley.

We derive this torque from first principles: (i) force balance on an inclined vehicle, (ii) standard rolling resistance theory, (iii) quadratic aerodynamic drag, (iv) reflection through wheel radius and gear ratio.

Step 1: Wheel and Gear Reflection Let r_w denote effective wheel rolling radius, and $G > 0$ the fixed gear ratio defined so that

$$\omega_s = G \omega_w.$$

Vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G} \omega_s(t). \quad (10.4)$$

If $F_{\text{road}}(t)$ denotes the total longitudinal resistive force on the vehicle, then the wheel torque is

$$\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \quad (10.5)$$

Assuming ideal power transmission ([Assumption 12 \(Ideal CVT Power Transmission\)](#)), the reflected secondary torque is

$$\tau_{\text{load}}(t) = \frac{1}{G} \tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \quad (10.6)$$

Thus we now determine $F_{\text{road}}(t)$.

Step 2: Free-Body Diagram and Force Decomposition Consider the vehicle on an incline of angle α . Using [Theory 1 \(Newton's Second Law \(Translation\)\)](#), the longitudinal resistive force is modeled as

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \quad (10.7)$$

Diagram Placeholder (Incline Free-Body Diagram)

Insert a free-body diagram of the vehicle on an incline showing:

- Weight mg acting vertically downward,
- Decomposition into $mg \sin \alpha$ along the slope and $mg \cos \alpha$ normal to the slope,
- Normal force N perpendicular to the surface,
- Positive longitudinal direction $+x$,
- Vehicle velocity v along $+x$.

Step 3: Rolling Resistance from [Theory 14 \(Rolling Resistance\)](#), rolling resistance is modeled as proportional to the normal force:

$$F_{\text{rr}} = C_{\text{rr}} N. \quad (10.8)$$

From the incline free-body diagram, trigonometric decomposition of gravity gives

$$N = mg \cos \alpha. \quad (10.9)$$

Substituting [Equation \(10.9\)](#) into [Equation \(10.8\)](#):

$$F_{\text{rr}} = C_{\text{rr}} mg \cos \alpha. \quad (10.10)$$

Rolling resistance always opposes motion. To enforce directionality, we multiply by the sign of velocity:

$$F_{\text{rr}}(v, \alpha) = C_{\text{rr}} mg \cos \alpha \operatorname{sgn}(v). \quad (10.11)$$

If $v > 0$, the force acts opposite $+x$; if $v < 0$, it acts opposite $-x$.

Step 4: Grade Resistance The longitudinal gravitational component from the free-body diagram is

$$F_{\text{grade}}(\alpha) = mg \sin \alpha. \quad (10.12)$$

This term is positive for an uphill incline ($\alpha > 0$), meaning it resists forward motion.

Step 5: Aerodynamic Drag Following [Theory 13 \(Air Resistance \(Quadratic Drag\)\)](#), the aerodynamic force magnitude is

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A_f v^2. \quad (10.13)$$

To ensure drag opposes motion, we write

$$F_{\text{drag}}(v) = \frac{1}{2} \rho C_d A_f v^2 \text{sgn}(v). \quad (10.14)$$

Step 6: Total Road Force Substituting [Equation \(10.11\)](#), [Equation \(10.12\)](#), and [Equation \(10.14\)](#) into [Equation \(10.7\)](#) gives

$$\begin{aligned} F_{\text{road}}(t) &= C_{\text{rr}} mg \cos \alpha \text{sgn}(v(t)) \\ &\quad + mg \sin \alpha \\ &\quad + \frac{1}{2} \rho C_d A_f v(t)^2 \text{sgn}(v(t)). \end{aligned} \quad (10.15)$$

Using [Equation \(10.4\)](#), $v(t) = \frac{r_w}{G} \omega_s(t)$, the fully expanded load torque becomes

$$\tau_{\text{load}}(t) = \frac{r_w}{G} \left[C_{\text{rr}} mg \cos \alpha \text{sgn}\left(\frac{r_w}{G} \omega_s(t)\right) + mg \sin \alpha + \frac{1}{2} \rho C_d A_f \left(\frac{r_w}{G} \omega_s(t)\right)^2 \text{sgn}\left(\frac{r_w}{G} \omega_s(t)\right) \right]. \quad (10.16)$$

Equation [Equation \(10.16\)](#) is the explicit resisting torque used in [Equation \(10.2\)](#).

10.5 Inertia Definitions

The rotational equations [Equation \(10.1\)](#)–[Equation \(10.2\)](#) require the effective inertias about each pulley axis.

Primary Pulley Inertia The primary pulley is rigidly connected to the engine. Its effective inertia is therefore the direct sum of

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \quad (10.17)$$

where

- I_{eng} is the engine rotational inertia,
- I_{prim} is the primary CVT pulley inertia.

No reflection is required on the primary side since all components rotate at the same angular velocity ω_p .

Secondary Pulley Inertia All components downstream of the secondary pulley must be expressed as an equivalent inertia about the secondary pulley axis.

Let

- I_{sec} : secondary CVT pulley inertia,
- I_{gb} : gearbox + intermediate shaft inertia,
- I_{wheel} : wheel + hub inertia about the wheel axis,
- m : total vehicle mass,
- r_w : effective wheel rolling radius,
- G : fixed gear ratio between secondary pulley and wheel (defined so that $\omega_s = G\omega_w$).

Vehicle Mass Contribution The vehicle translational kinetic energy is given by [Theory 4 \(Kinetic Energy \(Translational\)\)](#):

$$T_{\text{trans}} = \frac{1}{2}mv^2,$$

with $v = r_w\omega_w$. Expressed in rotational form about the wheel axis:

$$T_{\text{trans}} = \frac{1}{2}(mr_w^2)\omega_w^2.$$

Thus the vehicle mass contributes an effective rotational inertia

$$I_{\text{veh,wheel}} = mr_w^2$$

about the wheel axis.

Reflection to the Secondary Pulley Because $\omega_s = G\omega_w$, we have $\omega_w = \omega_s/G$.

Expressing kinetic energy in terms of ω_s :

$$T = \frac{1}{2}I_{\text{wheel}}\left(\frac{\omega_s}{G}\right)^2 + \frac{1}{2}(mr_w^2)\left(\frac{\omega_s}{G}\right)^2.$$

Therefore inertias at the wheel are reflected to the secondary pulley by

$$I_{\text{reflected}} = \frac{I_{\text{wheel}} + mr_w^2}{G^2}.$$

Total Secondary Inertia The total effective inertia about the secondary pulley becomes

$$I_s = I_{\text{sec}} + I_{\text{gb}} + \frac{I_{\text{wheel}} + mr_w^2}{G^2}. \quad (10.18)$$

Equation [Equation \(10.18\)](#) defines the total inertia resisting angular acceleration of the secondary pulley.

10.6 Ratio Kinematics

The CVT ratio was defined geometrically in [Section 8.8](#) (see [Equation \(8.21\)](#)) as

$$R(t) = \frac{r_{s,\text{eff}}(t)}{r_{p,\text{eff}}(t)}.$$

Under ideal belt transmission, the pulley speeds satisfy

$$\omega_p(t) = R(t) \omega_s(t). \quad (10.19)$$

Differentiating [Equation \(10.19\)](#) *with respect to time* gives

$$\alpha_p(t) = R(t) \alpha_s(t) + \omega_s(t) \dot{R}(t), \quad (10.20)$$

where $\dot{R}(t) = \frac{dR}{dt}$ was derived previously (see [Equation \(8.49\)](#)).

Equation [Equation \(10.20\)](#) captures the coupling introduced by a time-varying ratio: even if $\alpha_s = 0$, a changing ratio ($\dot{R} \neq 0$) produces primary acceleration.

10.7 Torque Mapping Across the CVT

The belt transmits power from the primary pulley to the secondary pulley. In general,

$$\tau_p(t) \omega_p(t) = \tau_s(t) \omega_s(t) + P_{\text{loss}}(t), \quad (10.21)$$

where $P_{\text{loss}}(t)$ represents internal dissipation. This identity always holds; modeling assumptions determine how P_{loss} is represented.

Following [Assumption 12 \(Ideal CVT Power Transmission\)](#), we set $P_{\text{loss}}(t) = 0$, giving

$$\tau_p(t) \omega_p(t) = \tau_s(t) \omega_s(t). \quad (10.22)$$

Substituting [Equation \(10.19\)](#) yields

$$\tau_s(t) = R(t) \tau_p(t). \quad (10.23)$$

Equation [Equation \(10.23\)](#) is the explicit torque relation required to eliminate τ_s .

10.8 Eliminating Accelerations

We now have four equations (Equation (10.1), Equation (10.2), Equation (10.20), Equation (10.23)) and four unknowns (α_p , α_s , τ_p , τ_s). The goal is to eliminate the accelerations α_p and α_s to obtain a closed-form expression for the coupling torque τ_p in terms of the known quantities τ_{eng} , τ_{load} , R , and $\dot{R}$.

From Equation (10.1),

$$\tau_p = \tau_{\text{eng}} - I_p \alpha_p.$$

Substituting Equation (10.20):

$$\tau_p = \tau_{\text{eng}} - I_p (R \alpha_s + \omega_s \dot{R}). \quad (10.24)$$

From Equation (10.2) and Equation (10.23),

$$\alpha_s = \frac{R \tau_p - \tau_{\text{load}}}{I_s}.$$

Substituting into Equation (10.24):

$$\begin{aligned} \tau_p &= \tau_{\text{eng}} - I_p \left[R \left(\frac{R \tau_p - \tau_{\text{load}}}{I_s} \right) + \omega_s \dot{R} \right] \\ &= \tau_{\text{eng}} - \frac{I_p}{I_s} (R^2 \tau_p - R \tau_{\text{load}}) - I_p \omega_s \dot{R}. \end{aligned} \quad (10.25)$$

Collecting terms:

$$\tau_p \left(1 + \frac{I_p}{I_s} R^2 \right) = \tau_{\text{eng}} + \frac{I_p}{I_s} R \tau_{\text{load}} - I_p \omega_s \dot{R}. \quad (10.26)$$

10.9 Closed-Form Coupling Torque

Solving Equation (10.26):

Coupling Torque (Primary Side) (10.27) $\tau_p(t) = \frac{\tau_{\text{eng}}(t) + \frac{I_p}{I_s} R(t) \tau_{\text{load}}(t) - I_p \omega_s(t) \dot{R}(t)}{1 + \frac{I_p}{I_s} R(t)^2}.$

The coupling torque as seen at the secondary pulley, required by Equation (9.15), then follows from Equation (10.23):

$$\tau_s(t) = R(t) \tau_p(t).$$

10.10 Section Summary

We began with rotational equations of motion for the primary and secondary pulleys. These introduced more unknowns than equations. Closure was obtained by incorporating:

- The geometric ratio definition $R(t)$,
- Its time derivative $\dot{R}(t)$,
- The torque mapping implied by power transfer across the CVT.

The resulting coupling torque [Equation \(10.27\)](#) expresses the torque transmitted through the CVT in terms of the instantaneous ratio and ratio rate. Substitution of [Equation \(10.28\)](#) into [Equation \(9.15\)](#) completes the dynamical coupling required for the axial shift equation [Equation \(9.35\)](#).

11 Traction Limit and Slip Condition

The torque transmission law derived previously assumes ideal no-slip belt traction. Under that assumption, the coupling torque [Equation \(10.27\)](#) (and equivalently [Equation \(10.28\)](#)) represents the torque actually transmitted through the CVT.

In reality, belt–pulley friction is finite. For a given normal clamping force, there exists a maximum tangential traction force that the contact can sustain. If the demanded coupling torque exceeds this traction capacity, slip occurs and the transmitted torque saturates at a friction-limited value.

The objective of this section is to derive a traction inequality of the form

$$|\tau| \leq \tau_{\max}, \tag{11.1}$$

which will later be enforced in place of the ideal coupling law whenever the no-slip solution violates this bound.

11.1 Differential Belt Element and Tangential Force Balance

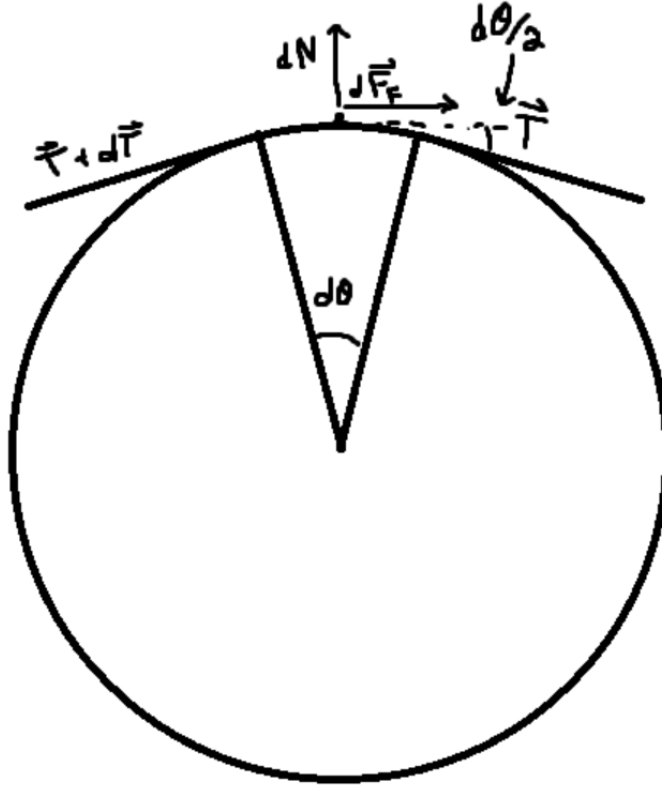


Figure 12: Free-body diagram of a differential belt element spanning $d\theta$.

Consider a differential belt element spanning an infinitesimal wrap angle $d\theta$ on a pulley, as shown in [Figure 12](#). The belt tension entering the element is T , and leaving the element is $T + dT$.

The pulley exerts on the belt element:

- a differential normal force dN (radial),
- a differential friction force dF_f (tangential).

The positive tangential direction is taken along the midpoint tangent of the element, consistent with increasing wrap angle.

Tangential Newton's Second Law Applying [Theory 1 \(Newton's Second Law \(Translation\)\)](#) in the tangential direction:

$$\sum F_t = dm a_t, \quad (11.2)$$

where a_t is the tangential acceleration of the belt material at the contact point.

The tangential components of the end tensions follow directly from geometry:

$$(T)_t = T \cos\left(\frac{d\theta}{2}\right), \quad (11.3)$$

$$(T + dT)_t = (T + dT) \cos\left(\frac{d\theta}{2}\right). \quad (11.4)$$

The left-end tension acts in the positive tangential direction, while the right-end tension acts in the negative tangential direction. Including friction, the tangential force balance becomes

$$T \cos\left(\frac{d\theta}{2}\right) - (T + dT) \cos\left(\frac{d\theta}{2}\right) + dF_f = dm a_t. \quad (11.5)$$

Simplifying the first two terms exactly,

$$-dT \cos\left(\frac{d\theta}{2}\right) + dF_f = dm a_t. \quad (11.6)$$

Rearranging,

$$dT \cos\left(\frac{d\theta}{2}\right) = dF_f - dm a_t. \quad (11.7)$$

Belt mass element From Equation (9.23),

$$dm = \mu_b r_{\text{cm}} d\theta, \quad (11.8)$$

so

$$dm a_t = \mu_b r_{\text{cm}} a_t d\theta. \quad (11.9)$$

Substituting into Equation (11.7),

$$dT \cos\left(\frac{d\theta}{2}\right) = dF_f - \mu_b r_{\text{cm}} a_t d\theta. \quad (11.10)$$

Friction limit and impending slip Theory 10 (Coulomb Friction) states that the friction force magnitude is bounded by the normal force:

$$|dF_f| \leq \mu dN. \quad (11.11)$$

The maximum transmissible tension increase occurs when friction is fully mobilized in the direction that increases tension (impending slip):

$$dF_f = +\mu dN. \quad (11.12)$$

Substituting Equation (11.12) into Equation (11.10) gives the maximum local tension change:

$$dT_{\max} = \frac{\mu dN - \mu_b r_{\text{cm}} a_t d\theta}{\cos\left(\frac{d\theta}{2}\right)}. \quad (11.13)$$

11.2 Integration Across the Wrap

Define the tight-slack tension difference

$$\Delta T = T_{\text{tight}} - T_{\text{slack}} = \int_0^\phi dT. \quad (11.14)$$

The maximum admissible tension difference is therefore

$$\Delta T_{\max} = \int_0^\phi \frac{\mu dN - \mu_b r_{\text{cm}} a_t d\theta}{\cos\left(\frac{d\theta}{2}\right)}. \quad (11.15)$$

In the differential limit $d\theta \rightarrow 0$, $\cos(d\theta/2) \rightarrow 1$, yielding

$$\Delta T_{\max} = \int_0^\phi \mu dN - \int_0^\phi \mu_b r_{\text{cm}} a_t d\theta. \quad (11.16)$$

Define total normal force over the wrap:

$$N_\phi = \int_0^\phi dN. \quad (11.17)$$

Assuming r_{cm} and a_t are approximately constant over the wrap,

$$\Delta T_{\max} = \mu N_\phi - \mu_b r_{\text{cm}} a_t \phi. \quad (11.18)$$

11.3 Torque Capacity Inequality

The transmitted torque magnitude is

$$|\tau| = r_{\text{eff}} |\Delta T|. \quad (11.19)$$

Substituting Equation (11.18) gives the traction limit

$$|\tau| \leq r_{\text{eff}} (\mu N_\phi - \mu_b r_{\text{cm}} a_t \phi). \quad (11.20)$$

Equation (11.20) provides the general traction-limited torque bound. However, two quantities remain to be defined: the normal force per unit arc length N_ϕ and the tangential acceleration a_t . In the next subsection, a_t will be expressed explicitly in terms of pulley angular acceleration and radial rate.

11.4 Tangential Acceleration of a Belt Element on a Circular Wrap

This subsection derives the tangential acceleration term required in the local traction inequality in §??. In that inequality we apply Newton's second law *along the direction of belt motion*, so we need an explicit expression for the belt element's tangential acceleration a_t .

Goal For a belt element of differential mass dm moving along the wrap, the tangential force balance uses

$$\sum F_t = dm a_t,$$

so our goal is to compute a_t from the belt kinematics when both *angular speed* and *wrap radius* may vary:

$$\omega(t) = \dot{\theta}(t), \quad \alpha(t) = \dot{\omega}(t), \quad r(t), \quad \dot{r}(t).$$

Coordinate system and unit vectors Because we care about forces *along the belt*, we choose a basis aligned with the belt's local geometry:

- $\mathbf{e}_r(t)$ points radially outward from the pulley center O to the belt point P ,
- $\mathbf{e}_\theta(t)$ is the unit tangent, obtained by rotating $\mathbf{e}_r$ by $+90^\circ$ in the direction of increasing θ .

These are not arbitrary symbols: they are *the* directions of radial and tangential motion on a circle, and the belt's local tangential direction is exactly $\mathbf{e}_\theta$.

Placeholder: polar basis at the wrap. Draw pulley center O and belt point P at radius r and angle θ . Show $\mathbf{e}_r$ pointing from O to P and $\mathbf{e}_\theta$ perpendicular to $\mathbf{e}_r$ in the direction of belt motion (increasing θ). Label belt tangential direction as $\mathbf{e}_\theta$.

Figure 13: Definition of the radial and tangential unit vectors.

Work backwards: to get a_t , compute $\mathbf{a}$ and take its tangential component Tangential acceleration is the component of the acceleration vector along $\mathbf{e}_\theta$:

$$a_t \equiv \mathbf{a} \cdot \mathbf{e}_\theta. \quad (11.21)$$

So we will compute $\mathbf{a} = \dot{\mathbf{v}}$, then read off the $\mathbf{e}_\theta$ component.

Step 1: position and velocity The position of P from the center is

$$\mathbf{r}(t) = r(t) \mathbf{e}_r(t). \quad (11.22)$$

Differentiate **Equation (11.22)** using the product rule:

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r} \mathbf{e}_r + r \dot{\mathbf{e}}_r. \quad (11.23)$$

This shows why we must track how the unit vectors change with time: even if r were constant, $\mathbf{e}_r$ rotates as the belt moves around the pulley.

Step 2: derive $\dot{\mathbf{e}}_r$ and $\dot{\mathbf{e}}_\theta$ (no assumptions) Over a small change $d\theta$, the unit vectors rotate rigidly by the same angle. The derivative with respect to angle is therefore:

$$\frac{d\mathbf{e}_r}{d\theta} = \mathbf{e}_\theta, \quad \frac{d\mathbf{e}_\theta}{d\theta} = -\mathbf{e}_r. \quad (11.24)$$

(Geometric justification: increasing θ rotates $\mathbf{e}_r$ toward the tangent; increasing θ rotates the tangent toward $-\mathbf{e}_r$.)

Convert angle-derivatives to time-derivatives with the chain rule $d(\cdot)/dt = (d(\cdot)/d\theta)\dot{\theta}$ and $\omega = \dot{\theta}$:

$$\dot{\mathbf{e}}_r = \omega \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\omega \mathbf{e}_r. \quad (11.25)$$

Step 3: velocity in radial–tangential form Substitute $\dot{\mathbf{e}}_r$ from Equation (11.25) into Equation (11.23):

$$\mathbf{v} = \dot{r} \mathbf{e}_r + r\omega \mathbf{e}_\theta. \quad (11.26)$$

This is already intuitive: $\dot{r}$ is the radial speed, and $r\omega$ is the tangential speed.

Step 4: acceleration and its tangential component Differentiate Equation (11.26):

$$\begin{aligned} \mathbf{a} = \dot{\mathbf{v}} &= \frac{d}{dt}(\dot{r} \mathbf{e}_r) + \frac{d}{dt}(r\omega \mathbf{e}_\theta) \\ &= \ddot{r} \mathbf{e}_r + \dot{r} \dot{\mathbf{e}}_r + (\dot{r}\omega + r\alpha) \mathbf{e}_\theta + r\omega \dot{\mathbf{e}}_\theta. \end{aligned} \quad (11.27)$$

Now substitute the unit-vector time derivatives from Equation (11.25):

$$\dot{\mathbf{e}}_r = \omega \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\omega \mathbf{e}_r,$$

and collect like directions:

$$\begin{aligned} \mathbf{a} &= \ddot{r} \mathbf{e}_r + \dot{r}(\omega \mathbf{e}_\theta) + (\dot{r}\omega + r\alpha) \mathbf{e}_\theta + r\omega(-\omega \mathbf{e}_r) \\ &= (\ddot{r} - r\omega^2) \mathbf{e}_r + (r\alpha + 2\dot{r}\omega) \mathbf{e}_\theta. \end{aligned} \quad (11.28)$$

Finally, by the definition Equation (11.21), the tangential acceleration is the $\mathbf{e}_\theta$ component:

$$a_t = r\alpha + 2\dot{r}\omega. \quad (11.29)$$

Where the factor of 2 comes from The two $\dot{r}\omega$ contributions in Equation (11.28) come from *two distinct effects*:

- $\dot{r} \dot{\mathbf{e}}_r$: even the *radial* velocity $\dot{r} \mathbf{e}_r$ gains a tangential contribution because $\mathbf{e}_r$ rotates at rate ω ;
- $\frac{d}{dt}(r\omega)$: the tangential speed $r\omega$ changes if r changes.

Adding these yields the total $2\dot{r}\omega$ term.

11.5 Normal Force from Axial Clamping

The local traction inequality derived in ?? depends on the normal load over the wrap,

$$N_\phi \equiv \int_0^\phi dN,$$

since the available friction is μdN at each differential contact element.

In the CVT, this normal load is not generated implicitly by belt tension (as in the classical capstan problem). Instead, it is imposed by the pulley *axial clamping force* derived in Section 9. We now map axial clamp into an equivalent total normal load over the wrap.

Geometry: why a factor of $2 \tan \beta$ appears Recall the conical kinematics established earlier for opposing sheaves (Section 8.3), where axial closure beyond the deadzone produces a radial change

$$\Delta r = \frac{\Delta s}{2 \tan \beta} \quad \implies \quad \frac{dr}{ds} = \frac{1}{2 \tan \beta}.$$

This is the key geometric ratio: a given axial displacement ds produces a smaller radial displacement dr by the factor $2 \tan \beta$, because the closure is shared between the two opposing faces.

To convert forces consistently, we use virtual work (equivalently, power balance in the shift direction): the incremental work done by the axial clamp must equal the incremental work associated with the corresponding radial normal loading,

$$F_{\text{ax}} ds = F_{\text{rad}} dr.$$

Substituting $dr = (dr/ds) ds$ gives

$$\begin{aligned} F_{\text{ax}} ds &= F_{\text{rad}} \left(\frac{dr}{ds} \right) ds \\ \implies F_{\text{rad}} &= \frac{F_{\text{ax}}}{dr/ds} = F_{\text{ax}} (2 \tan \beta). \end{aligned} \tag{11.30}$$

Thus the total radial normal load supplied by a given axial clamp force is

$$F_{\text{rad}} = 2 \tan \beta F_{\text{ax}}. \tag{11.31}$$

Identification with wrap-normal load N_ϕ Under the uniform distribution assumption (discussed previously), we take the total normal load over the wrap to be equal to the total radial normal load supplied by clamping:

$$N_\phi = F_{\text{rad}} = 2 \tan \beta F_{\text{ax}}. \tag{11.32}$$

Equivalently, the normal force per unit wrap angle is constant:

$$\frac{dN}{d\theta} = \frac{N_\phi}{\phi} \quad \implies \quad dN = \frac{N_\phi}{\phi} d\theta. \tag{11.33}$$

Substituting Equation (11.32) into the integrated traction bound (from ??) yields the clamp-limited torque capacity form

$$|\tau| \leq r_{\text{eff}} [\mu N_\phi - \mu_b r_{\text{cm}} a_t \phi] = r_{\text{eff}} [\mu (2 \tan \beta F_{\text{ax}}) - \mu_b r_{\text{cm}} a_t \phi]. \quad (11.34)$$

Remark (Why this differs from the classical capstan exponential)

In the classical capstan problem, the normal load is generated *by the belt tension* itself (a larger tension implies a proportionally larger contact normal), which leads to the exponential relation $T_1/T_2 = e^{\mu\phi}$.

Here, the normal load is imposed externally by the clamping mechanism: N_ϕ is determined by F_{ax} through Equation (11.32). Therefore, the available traction scales *linearly* with clamping force, not exponentially with belt tension. Centrifugal loading and ratio change enter additionally through the inertial term in Equation (11.34).

11.6 Section Conclusion and Transition

This section derived a traction-limited torque bound for a single pulley.

The result is

$$|\tau| \leq r_{\text{eff}} [\mu (2 \tan \beta F_{\text{ax}}) - \mu_b r_{\text{cm}} a_t \phi], \quad (11.35)$$

with

$$a_t = r_{\text{cm}} \alpha + 2 \dot{r}_{\text{cm}} \omega.$$

Equation Equation (11.35) gives the maximum torque that can be transmitted at the current operating state. It depends only on:

- axial clamping force F_{ax} ,
- cone angle β ,
- wrap angle ϕ ,
- belt mass and kinematics.

No exponential amplification occurs because the normal force is externally imposed and bounded.

The ideal coupling torque derived earlier remains valid only when it satisfies Equation (11.35). If the demanded torque exceeds this bound, the belt cannot sustain static friction and slip must occur.

The next section defines the stick and slip regimes and modifies the governing equations accordingly.

12 Slip Regimes and Traction-Limited Coupling Torque

Placeholder: Slip intuition diagram.

Sketch two pulleys with effective radii $r_{p,\text{eff}}$, $r_{s,\text{eff}}$ and angular speeds ω_p, ω_s . Show the two tangential surface speeds $v_p = r_{p,\text{eff}}\omega_p$ and $v_s = r_{s,\text{eff}}\omega_s$ at the belt contact, and define $v_{\text{rel}} = v_p - v_s$. Annotate: $v_{\text{rel}} = 0$ (stick) vs $v_{\text{rel}} \neq 0$ (slip), and indicate the friction torque direction opposing v_{rel} .

Figure 14: Conceptual definition of slip via contact-speed mismatch.

12.1 Working Backwards From the Goal

The goal of this section is to define, at every instant, a *physically admissible* coupling torque that the belt can transmit between the pulleys. Earlier, we derived:

- a no-slip (rigid-traction) relationship that specifies how torque transfers between the pulleys when the belt adheres (“stick”), and
- a traction capacity bound for each pulley of the form $|\tau| \leq \tau_{\text{max},p}$ and $|\tau| \leq \tau_{\text{max},s}$.

These two results must be combined into a single rule that:

1. *never* violates either capacity bound (even before slip begins),
2. reduces to the earlier no-slip torque transfer when feasible, and
3. produces a slip torque that opposes relative motion when the no-slip demand is infeasible.

To do this cleanly, we proceed in the order that matches physical intuition:

(i) *define how we detect slip* $\rightarrow$ (ii) *define the torque limits that force entry into slip* $\rightarrow$ (iii) *define the torque law during slip* $\rightarrow$ (iv) *define how we exit slip*.

12.2 Slip Metric: How We Know We Are Slipping

Slip in a belt CVT is most naturally defined at the belt–pulley contact: if the belt adheres to both pulleys, then the tangential surface speeds at the contact must match. Using the effective contact radii, define

$$v_p \triangleq r_{p,\text{eff}} \omega_p, \quad v_s \triangleq r_{s,\text{eff}} \omega_s, \quad (12.1)$$

and the relative contact-speed mismatch

$$v_{\text{rel}} \triangleq v_p - v_s = r_{p,\text{eff}}\omega_p - r_{s,\text{eff}}\omega_s. \quad (12.2)$$

Stick condition (no slip):

$$v_{\text{rel}} = 0. \quad (12.3)$$

Slip condition:

$$v_{\text{rel}} \neq 0. \quad (12.4)$$

Equation [Equation \(12.2\)](#) is therefore the diagnostic: it is the quantity we monitor to determine whether the belt is adhering (stick) or sliding (slip). In a perfect rigid-traction model, the dynamics

would enforce Equation (12.3) exactly. In a traction-limited model, Equation (12.3) can only be maintained if the required torques remain within the available traction capacity.

12.3 Traction Capacity: How We Enter Slip

From the traction analysis of the previous section, each pulley has a maximum transmissible torque magnitude at the current operating state:

$$|\tau_p| \leq \tau_{\max,p}(\cdot), \quad |\tau_s| \leq \tau_{\max,s}(\cdot), \quad (12.5)$$

where $\tau_{\max,p}$ and $\tau_{\max,s}$ depend on the current state (clamping force, cone angle, wrap angle, belt kinematics, etc.) through Equation (11.35) applied to the primary and secondary, respectively.

Key physical point: these bounds apply *always*. Even before any slip is observed, the belt–pulley interface cannot transmit a torque exceeding its instantaneous capacity. Therefore, if the no-slip model (rigid traction) would *demand* a torque larger than the bound in Equation (12.5), the constraint $v_{\text{rel}} = 0$ can no longer be maintained by static friction.

In practice, torque transfer is constrained by the *most limiting* interface. We therefore define a single admissible coupling capacity

$$\tau_{\max} \triangleq \min(\tau_{\max,p}, \tau_{\max,s}). \quad (12.6)$$

Intuition for entering slip. Suppose the system is in stick so that $v_{\text{rel}} = 0$ initially. External conditions (engine torque rise, vehicle load increase, ratio change reducing effective radius leverage, reduced clamping, etc.) may increase the torque required to preserve $v_{\text{rel}} = 0$. As soon as the required coupling torque exceeds $\tau_{\max}$, the transmitted torque cannot increase further. The coupling torque saturates at the traction limit, while the remaining unbalanced torque accelerates one pulley relative to the other, causing Equation (12.3) to break and v_{rel} to grow in magnitude. This is the mechanism by which the model *falls into* a slip regime.

12.4 Coupling Torque Law: Stick When Feasible, Slip When Necessary

Earlier sections already specify how torques transfer *when not slipping*: the dynamics and kinematics provide a “desired” (no-slip) coupling torque that would maintain $v_{\text{rel}} = 0$. We denote this no-slip demand by

$$\tau_{\text{ns}} \triangleq (\text{the coupling torque predicted by the no-slip torque-transfer model}). \quad (12.7)$$

This quantity is not a new model; it is simply a name for “whatever torque the earlier no-slip equations imply” at the current state.

However, the belt cannot transmit an arbitrarily large torque: it must also satisfy Equation (12.6). Therefore, even in stick, the admissible coupling torque is at most $\tau_{\max}$ in magnitude.

(A) Feasible no-slip torque (stick). If the no-slip demand is within capacity,

$$|\tau_{\text{ns}}| \leq \tau_{\max}, \quad (12.8)$$

then static friction can support the demand and the belt remains in stick:

$$\tau_c = \tau_{\text{ns}}, \quad v_{\text{rel}} = 0. \quad (12.9)$$

(B) Infeasible no-slip torque (entering slip). If the no-slip demand exceeds capacity,

$$|\tau_{\text{ns}}| > \tau_{\text{max}}, \quad (12.10)$$

then the coupling torque must saturate and the mismatch v_{rel} can no longer remain zero. In slip, the transmitted torque is friction-limited and its direction must oppose the relative motion. A smooth Coulomb-like slip law is

$$\tau_{\text{slip}} = \tau_{\text{max}} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \quad (12.11)$$

where $v_{\text{smooth}} > 0$ controls the transition sharpness near $v_{\text{rel}} = 0$. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches $\tau_{\text{max}} \text{sign}(v_{\text{rel}})$.

Sign convention note. Equation [Equation \(12.11\)](#) is written so that the slip torque changes sign with v_{rel} . In the equations of motion, τ_c must be applied with opposite sign on the two pulleys (equal-and-opposite action), so that the net effect is always to *reduce* the mismatch and dissipate energy during slip. (If your earlier rotational EOM use the opposite sign convention for τ_c , the tanh argument should be negated accordingly.)

(C) Smooth unified law (recommended for numerics). Rather than hard switching, we define a blending factor $\alpha \in [0, 1]$ that increases from stick-like to slip-like behavior as the relative speed grows:

$$\alpha(v_{\text{rel}}) = \text{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}}, 0, 1\right), \quad (12.12)$$

where $v_{\text{blend}} > 0$ is a chosen blending scale. We also define the capacity-saturated no-slip torque

$$\tau_{\text{ns,sat}} = \text{clip}(\tau_{\text{ns}}, -\tau_{\text{max}}, +\tau_{\text{max}}). \quad (12.13)$$

The final coupling torque is then

$$\tau_c = (1 - \alpha) \tau_{\text{ns,sat}} + \alpha \tau_{\text{slip}}, \quad (12.14)$$

with τ_{slip} given by [Equation \(12.11\)](#).

This construction encodes all desired behaviors:

- **Capacity is never violated:** both $\tau_{\text{ns,sat}}$ and τ_{slip} are bounded by τ_{max} , hence so is τ_c .
- **Stick when feasible:** if $v_{\text{rel}} \approx 0$, then $\alpha \approx 0$ and $\tau_c \approx \tau_{\text{ns,sat}}$ (equal to τ_{ns} if feasible).
- **Slip when necessary:** if mismatch grows, $\alpha \rightarrow 1$ and $\tau_c \rightarrow \tau_{\text{slip}}$, i.e. friction-limited sliding torque.

12.5 How Slip Reduces and Ends: Exiting the Slip Regime

Once the coupling torque saturates, the remainder of the external torque imbalance accelerates the pulleys differently, which produces a nonzero v_{rel} through [Equation \(12.2\)](#). During slip, the transmitted torque magnitude cannot exceed τ_{max} , but its *direction* always opposes the mismatch via [Equation \(12.11\)](#). As a result, the slip torque acts to reduce $|v_{\text{rel}}|$ whenever the external conditions permit.

Intuition for exiting slip. Slip persists as long as the system would require more than τ_{max} to enforce $v_{\text{rel}} = 0$. If operating conditions change (e.g., demanded torque falls, clamping increases,

wrap angle increases, ratio shifts into a more favorable radius, or the inertial term reduces), then $\tau_{\max}$ may rise and/or the required no-slip demand τ_{ns} may fall. When the inequality becomes feasible again,

$$|\tau_{\text{ns}}| \leq \tau_{\max}, \quad (12.15)$$

static traction can re-establish adherence and the mismatch decays to zero. In the smooth blended law [Equation \(12.14\)](#), this corresponds to $v_{\text{rel}} \rightarrow 0$, so that $\alpha \rightarrow 0$ and $\tau_c \rightarrow \tau_{\text{ns}}$ continuously.

12.6 Implementation Summary (Matches the Simulation Rule)

For convenience, the continuous-time model above corresponds directly to the following computation at each time step:

1. compute $v_{\text{rel}} = r_{p,\text{eff}}\omega_p - r_{s,\text{eff}}\omega_s$,
2. compute capacities $\tau_{\max,p}$ and $\tau_{\max,s}$, then $\tau_{\max} = \min(\tau_{\max,p}, \tau_{\max,s})$,
3. compute the no-slip demand τ_{ns} from the earlier torque-transfer equations,
4. form $\tau_{\text{ns,sat}} = \text{clip}(\tau_{\text{ns}}, -\tau_{\max}, +\tau_{\max})$,
5. form $\tau_{\text{slip}} = \tau_{\max} \tanh(v_{\text{rel}}/v_{\text{smooth}})$,
6. blend using $\alpha = \text{clip}(|v_{\text{rel}}|/v_{\text{blend}}, 0, 1)$: $\tau_c = (1 - \alpha)\tau_{\text{ns,sat}} + \alpha\tau_{\text{slip}}$.

This yields a coupling torque that (i) respects traction limits at both pulleys, (ii) reproduces the earlier no-slip behavior when feasible, and (iii) transitions smoothly into a Coulomb-like slip regime when necessary.

A Parameter Tables

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